

Photoinduced anisotropic edge states and signatures of nonequilibrium spin polarization in silicene nanoribbons

Jia-kun Wang^a, Jia-Yu Li^{a,b}, Hou-Jian Duan^a, Rui-Qiang Wang^a

^a Guangdong Provincial Key Laboratory of Quantum Engineering and Quantum Materials, and School of Physics and Telecommunication Engineering, South China Normal University, Guangzhou 510006, China

^b Shenzhen Institute for Quantum Science and Technology, and Department of Physics, Southern University of Science and Technology, Shenzhen, 518055, China

ARTICLE INFO

Article history:

Received 5 January 2020

Received in revised form 10 March 2020

Accepted 23 March 2020

Available online 26 March 2020

Communicated by L. Ghivelder

Keywords:

Silicene

Spin polarization

Off-resonant light

Topological phase

ABSTRACT

Most topological phase transitions are accompanied by the emergence of surface/edge states with spin dependence. Usually, the quantized Hall conductivity cannot characterize the anisotropic transports and spin dependence of topological states. Here, we study the intricate topological phase transition and the anisotropic behavior of edge states in silicene nanoribbon submitted to an electric field or/and a light irradiation. It is interesting to find that a circularly polarized light can induce a type-II quantum anomaly Hall phase, which is manifested as the high Chern number and the strong anisotropic edge states. Besides the measurement of the quantized Hall conductivity, we further propose to probe these topological phase transitions and the anisotropy of edge states by measuring the current-induced nonequilibrium spin polarization. It is found that the spin polarization exhibits more signatures about the behavior of surface/edge states, beyond the quantized Hall conductivity, especially for spin-dependent transports with different velocities.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Inverse spin galvanic effect (ISGE), referring to the electrical or optical generation of a nonequilibrium spin polarization in non-magnetic materials, e.g., those with spin-orbit interactions, has attracted much attention theoretically as well as experimentally over the years [1]. The ISGE is an important origin of spin torque [2], which can in turn switch the magnetization of a proximate ferromagnetic layer, and thus it is crucial to electrically generate the spin polarization. With the advancement of abundant topological materials, topological phase transitions provide a new way to control the ISGE, or inversely we can extract the message of topological phases from the probe of the ISGE. Therefore, study on the relation between topological phases and spin polarization becomes very important.

Silicene, a two-dimensional topological insulator constructed by a single layer of silicon atoms [3], synthesised in recent experiments [4–6], offers an ideal platform for exploring the phase transition. Different from graphene, silicene possesses large spin-orbit interaction (SOI) [7–12] and a buckled hexagonal structure

(two atoms in the translational unit cell reside on different planes) [13]. The former results in quantum spin Hall (QSH) effect [14,15] whereas the latter makes its band gap tunable easily by applying an electric field perpendicular to the silicene sheet [16–19]. As a result, one can externally tune the band gap of silicene and drive topological phase transitions from the QSH insulator to other intricate phases [12,14,15,20,21]. Meanwhile, the periodically driven topological properties have also been studied in other two dimensional materials, including graphene [22–24], semiconductor quantum wells [25], topological insulators [26,27] and other hexagonal lattice models [28]. Among them, an open issue is how to detect these intricate phase transitions experimentally. Jin et al. [29] have suggested to measure the bulk Nernst conductivity, from which the phase boundaries can be determined by comparing the charge-with spin-Nernst conductivities. Ezawa et al. have proposed to observe the phase transition points by measuring diamagnetism [30], optical absorption [20], or density of states [8]. Our previous work [31] has found that the bulk RKKY measurement can provide significant signatures to identify different phases and phase boundaries simultaneously.

Most topological phase transitions are accompanied by the emergence of surface/edge states with spin dependence. Different from the measurement of bulk bands, the most instinctive method to detect a topological phase is to directly measure the

E-mail addresses: dhjphd@163.com (H.-J. Duan), rqwanggz@163.com (R.-Q. Wang).

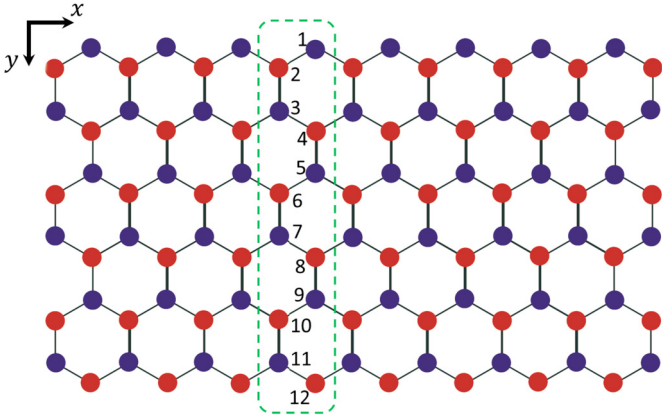


Fig. 1. (Color online.) The top view of the zigzag silicene nanoribbons, where the blue (red) circle denotes for A (B) sublattice. The translational unit cell is highlighted by the dashed box. A beam of circularly polarized light and an external electric field are exerted normally to the surface of the nanoribbon.

quantized Hall conductivity or observing surface/edge states [20]. Usually, the quantized Hall conductivity cannot characterize the spin dependence of topological states, and specially cannot reflect their anisotropic transports. One can notice that when charges flow at the surface/edge of topological insulators, the accompanied nonequilibrium spin polarization appears. Recently, A. Dyrda et al. [32] and Li et al. [33] have explored the influence on nonequilibrium spin density of topological phase transition between the QSH and conventional band insulator (CBI), controlled by electric field. Topological phase transition in topological insulators can be controlled not only by an electric field but also by a light field. In this paper, we find that in the silicene nanoribbon, a light field can switch the QSH phase to type-II quantum anomaly Hall (QAH) phase which has high Chern number and strong anisotropic velocity. We further propose to probe these topological phase transitions and the anisotropy of edge states by measuring the spin polarization, beyond the quantized Hall conductance.

2. Model and method

We consider a single-layered zigzag silicene nanoribbon, which is truncated along the zigzag direction (x -axis) and finite in the armchair direction (y -axis). The corresponding huge translational unit cell is highlighted by the dashed box in Fig. 1, where two kinds of silicon atoms (red A and blue B) reside in different planes. This buckled structure provides opportunity to control its electronic structure by applying an electric field. In addition, a beam of circularly polarized light is assumed to be irradiated normal to the surface of the nanoribbon (along z -axis) with its electromagnetic potential described by $\mathbf{A}(t) = (A_0 \sin \Omega t, A_0 \cos \Omega t)$, where A_0 (Ω) denotes the amplitude (frequency) of the illumination. Subjected to above external fields, the effective tight-binding model for the silicene [15,21,34,35] reads as

$$H = -t \sum_{\langle i,j \rangle} c_{i\alpha}^\dagger c_{j\alpha} + i \frac{t_{s0}}{3\sqrt{3}} \sum_{\langle\langle i,j \rangle\rangle} v_{ij} c_{i\alpha}^\dagger (\sigma_z)_{\alpha\beta} c_{j\beta} - U \sum_{i\alpha} \mu_i c_{i\alpha}^\dagger c_{i\alpha} - \frac{t_\Omega}{3\sqrt{3}} \sum_{\langle\langle i,j \rangle\rangle} v_{ij} c_{i\alpha}^\dagger c_{j\beta}, \quad (1)$$

where $c_{i\alpha}^\dagger$ ($c_{i\alpha}$) annihilates (creates) an electron with spin α at site i , with $\langle i, j \rangle$ ($\langle\langle i, j \rangle\rangle$) running over the nearest-neighbor (the next-nearest-neighbor) hopping sites, σ is the vector of Pauli matrices in real spin space $\sigma = (\sigma_x, \sigma_y, \sigma_z)$. The first two terms describe the silicene with hopping energy $t = 1.6$ eV and the intrinsic spin-orbit coupling $\lambda_{s0} \approx 3.9$ meV. $v_{ij} = \pm 1$ label the anticlockwise and

clockwise hopping with respect to the positive z -axis [36,37]. The third term describes the staggered potential $U = E_z d/2$ due to an electric field E_z applied to two sublattices ($\mu_i = \pm 1$ stand for A(B) sublattice), separated by distance $d = 0.46$ Å. The last term represents the effect of the off-resonant circularly polarized light with illumination parameter $t_\Omega = \frac{3t^2 A^2}{4\hbar\Omega^2}$ [21,29,38] under the Haldane model [35]. This treatment is reasonable for the high-frequency limit ($\Omega > 3t/\hbar = 10^3$ THz) since it does not directly excite electrons but instead effectively modifies the band structure through the process of virtual photon absorption and emission [39]. In Eq. (1), we have also neglected the Rashba interactions since their effects are negligibly small [15].

Since the nanoribbon is terminated in the armchair direction (y -axis) with open boundary, the momentum k_y is no longer a good quantum number, and the system acts as a one-dimensional (1D) lattice with continuous momentum k_x . To describe the electronic structure of k_x -direction, a translational super unit cell is considered in the dashed box of Fig. 1, where the width of the nanoribbon is defined as N (N is the number of atoms in a unit cell). In the absence of the Rashba coupling, the Hamiltonian H of Eq. (1) becomes diagonal in spin space. For each spin ($s_z = \pm$ for spin-up and spin-down), by transforming the position space of x -axis into momentum space, the Hamiltonian in Eq. (1) can be rewritten as a spin-dependent $N \times N$ matrix $H_{s_z}(k_x)$, which reads as

$$H_{s_z}(k_x) = \begin{bmatrix} h_{s_z} & T & g_{s_z} & 0 & 0 & 0 & \cdots \\ T & -h_{s_z} & t & -g_{s_z} & 0 & 0 & \cdots \\ g_{s_z} & t & h_{s_z} & T & g_{s_z} & 0 & \cdots \\ 0 & -g_{s_z} & T & -h_{s_z} & t & -g_{s_z} & \cdots \\ 0 & 0 & g_{s_z} & t & h_{s_z} & T & \cdots \\ 0 & 0 & 0 & -g_{s_z} & T & -h_{s_z} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (2)$$

with matrix elements

$$h_{s_z} = s_z \frac{2}{3\sqrt{3}} t_{s0} \sin(k_x a) - \frac{2}{3\sqrt{3}} t_\Omega \sin(k_x a) + U, \\ T = 2t \cos\left(\frac{k_x a}{2}\right), \\ g_{s_z} = -s_z \frac{2}{3\sqrt{3}} t_{s0} \sin\left(\frac{k_x a}{2}\right) + \frac{2}{3\sqrt{3}} t_\Omega \sin\left(\frac{k_x a}{2}\right).$$

By diagonalizing the spin-dependent Hamiltonian $H_{s_z}(k_x)$, one can obtain the bulk bands, as well as the edge bands, for zigzag silicene nanoribbons, as shown in Fig. 2.

Within the framework of linear response theory of the Kubo formula [40–42], the spin polarization of the carriers can be calculated as

$$\delta \mathbf{S} = - \lim_{\omega \rightarrow 0} \frac{1}{\omega} \text{Re} \Pi_{\mathbf{s}, \mathbf{j}}(i\omega_n), \quad (3)$$

where $\Pi_{\mathbf{s}, \mathbf{j}}(i\omega_n)$ is the imaginary-time retarded correlation function between the spin $\mathbf{s} = \frac{\hbar}{2} \boldsymbol{\sigma}$ of conducting electron and current operator $\mathbf{j} = e\mathbf{v}$ with the velocity operator $\mathbf{v} = \frac{1}{\hbar} \nabla_{\mathbf{k}} H$. Considering analytical continuation $i\omega_n \rightarrow \omega + i0^+$, we obtain

$$\Pi_{\mathbf{s}, \mathbf{j}}(\Omega) = \sum_{\mathbf{k}} \int d\varepsilon f(\varepsilon) \text{Tr} \{ \mathbf{s} [G_{\mathbf{k}}^R(\varepsilon) - G_{\mathbf{k}}^A(\varepsilon)] (\mathbf{j} \cdot \mathbf{E}) G_{\mathbf{k}}^A(\varepsilon - \Omega) + \mathbf{s} G_{\mathbf{k}}^R(\varepsilon + \Omega) (\mathbf{j} \cdot \mathbf{E}) [G_{\mathbf{k}}^R(\varepsilon) - G_{\mathbf{k}}^A(\varepsilon)] \}. \quad (4)$$

Here, $\mathbf{E}$ is the electric field, $f(\varepsilon) = [e^{\beta(\varepsilon - \mu_F)} + 1]^{-1}$ is the Fermi-Dirac distribution function with the inverse of temperature $\beta =$

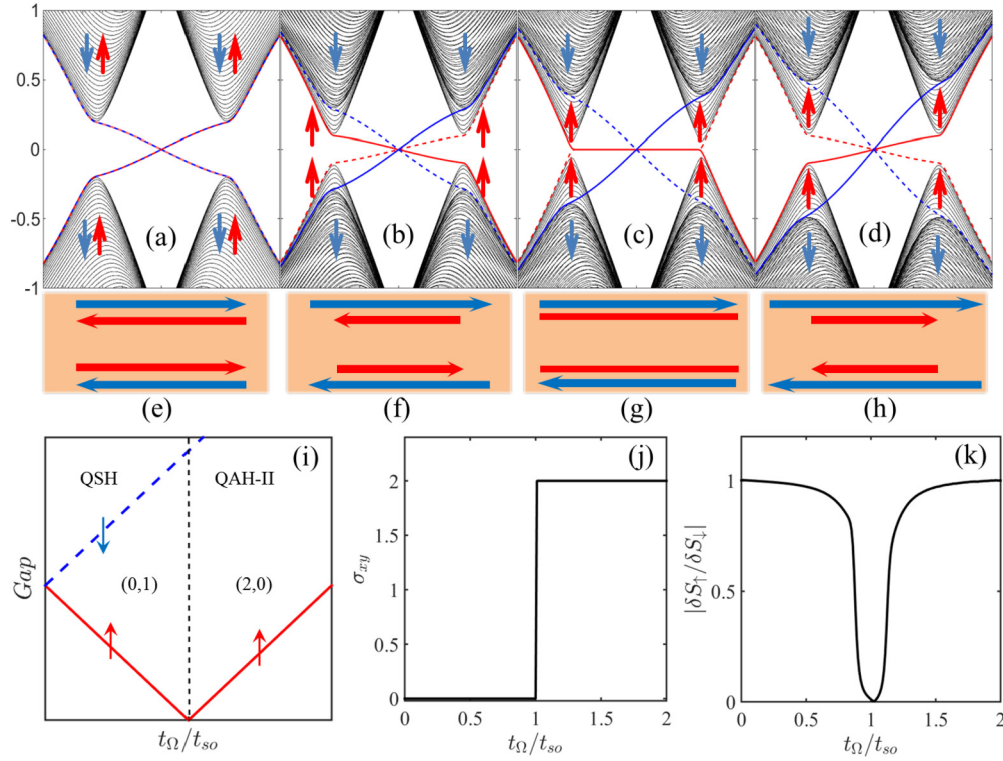


Fig. 2. (Color online.) (a)–(d) The evolution of photon-dressed band in a silicene nanoribbon with different light intensities ($t_\Omega = 0, 0.5, 1, 1.5$) and $U = 0$, in unit of t_{so} . The solid lines stand for up-edge state and the dashed lines for bottom-edge state. (e)–(h) Sketch map for edge states, corresponding to (a)–(d), respectively. The arrow direction denotes the transport direction and the arrow length stands for the velocity size. In (a)–(h), all red (blue) lines or arrows stand for spin-up (spin-down) electrons. (i) The phase transition QSH-QAH (type-II) with the light intensity t_Ω , where (C, C_s) labels with charge-Chern number C and spin-Chern number C_s . The variation of (j) the Hall conductivity σ_{xy} and (k) the ratio of spin polarization $|\delta S_\uparrow/\delta S_\downarrow|$ with t_Ω , respectively. We choose $u_F = 0.02$ eV.

$1/k_B T$ and chemical potential μ_F , and $G_{\mathbf{k}}^{R/A}(\varepsilon)$ is the retarded/advanced Green's function. The Green's function $G_{\mathbf{k}}^{R/A}(\varepsilon)$ can be obtained as

$$G_{\mathbf{k}}^{R/A}(\varepsilon) = \sum_n \frac{|\Phi_{\mathbf{k}n}\rangle \langle \Phi_{\mathbf{k}n}|}{\varepsilon - \varepsilon_{\mathbf{k}n} \pm i\Gamma}, \quad (5)$$

from the eigenvectors $|\Phi_{\mathbf{k}n}\rangle$ and eigenvalues $\varepsilon_{\mathbf{k}n}$ by diagonalizing Eq. (1), where n is the band index. Here, Γ is the small energy spectral broadening due to the weak impurity effect.

In presence of the intrinsic SOI, only the out-of-plane nonequilibrium spin polarization (δS_z) survives while the in-plane components ($\delta S_{x,y}$) are suppressed. At low temperature, one can reduce $\partial_\varepsilon f(\varepsilon) \rightarrow -\delta(E_F - \varepsilon)$, and the component δS_z mainly is contributed by the Fermi surface. Inserting $\Pi_{\mathbf{s},j}(\Omega)$ in Eq. (4) into Eq. (3) and taking $\Omega \rightarrow 0$, the spin polarization can be expressed as

$$\delta S_z = \frac{1}{2\pi A} \text{Re} \sum_{\mathbf{k}, nn'} \langle \Phi_{\mathbf{k}n} | \tau_0 \sigma_z | \Phi_{\mathbf{k}n'} \rangle \langle \Phi_{\mathbf{k}n'} | e\hbar \mathbf{E} \cdot \mathbf{v} | \Phi_{\mathbf{k}n} \rangle \times \frac{1}{\varepsilon - \varepsilon_{\mathbf{k}n} - i\Gamma} \left(\frac{1}{\varepsilon - \varepsilon_{\mathbf{k}n'} + i\Gamma} - \frac{1}{\varepsilon - \varepsilon_{\mathbf{k}n'} - i\Gamma} \right), \quad (6)$$

where A is the area of unit cell. Since the spin-polarized states embedded in the gap are mainly localized in the up (down) edge of nanoribbon, we focus on boundary spin polarization in our following discussion.

3. Results

The band structures of zigzag silicene nanoribbons calculated from Eq. (2) are shown in Fig. 2, where the edge bands of spin-up (spin-down) in the energy gap are marked by red (blue) lines. The

non-zero intrinsic SOI generates a bulk energy gap $\propto t_{so}$ and makes the system stay in a typical QSH state, as shown in Fig. 2 (a), where two kinds of spins are degenerated. Without the external fields, the time-reversal symmetry guarantees that at each boundary, the two spin-opposite edge states propagate in opposite direction with the same velocity (chiral edge states), whose sketch diagram is shown in Fig. 2(e).

Once a circularly polarized light is introduced, the time-reversal symmetry is broken. With the increase of the light intensity from Fig. 2(b) to Fig. 2(d), the circularly polarized light induces a monotonic increase of the bulk gap for spin-down band and a decrease for spin-up band. This in turn leads to two effects. One is that two spin-polarized edge channels, connecting the tips of the Dirac cones with the same spin between the K and K' valleys, have different slope or velocity, namely, anisotropic edge states. The other is that when the spin-down cone remains gap during the irradiation enhancement, the spin-up Dirac cones at both valleys K and K' are closed first at the critical point $t_\Omega = \pm t_{so}$ and then reopens a gap. The reopened gap inverses the bulk band structure of spin-up component, and thus reverses the sign of slope of the spin-up edge band. As a consequence, the two edge states with opposite spin propagate along the same direction, which implies that it enters the QAH states. Importantly, this QAH state is different from usual QAH state [43] due to the coexistence of two spin species with anisotropic slope. At the critical point $t_\Omega = \pm t_{so}$, the edge band of spin up becomes the topological flat band. We describe this dependence of edge states on radiation with sketch map in Figs. 2(e)–(h), where spin-up (spin-down) edge states are marked by red (blue) color and the arrow length stands for the velocity size. If the polarization direction of circularly polarized light is changed, the above results for two spin species exchange.

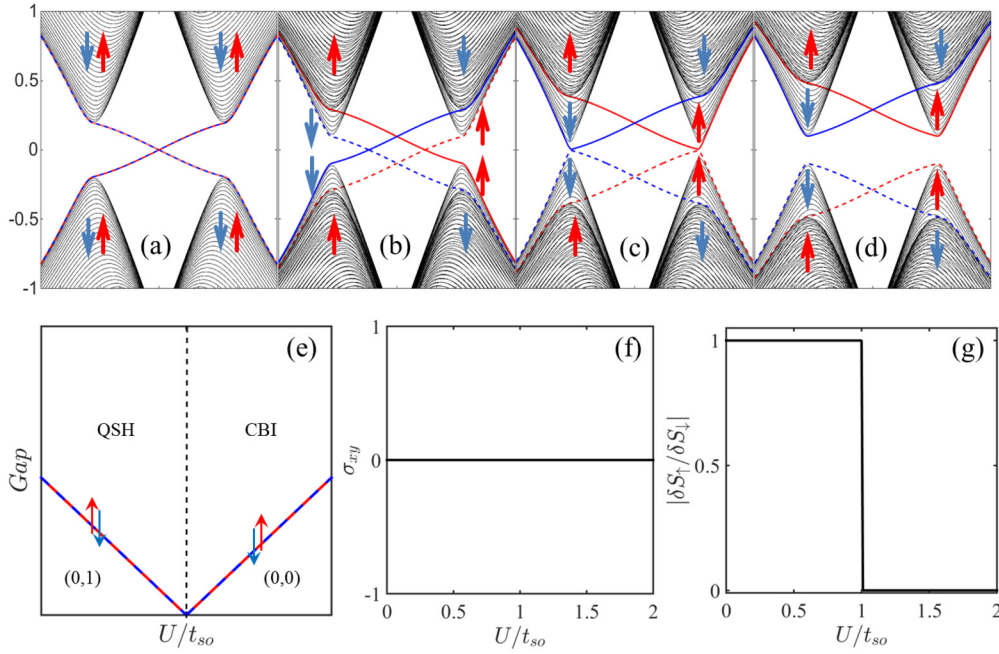


Fig. 3. (Color online.) (a)–(d) The evolution of energy band in a silicene nanoribbon with different electric fields ($U = 0, 0.5, 1, 1.5$) and $t_{\Omega} = 0$, in unit of t_{so} . The solid lines stand for up-edge state and the dashed lines for bottom-edge state. The red (blue) lines or arrows stand for spin-up (spin-down) electrons. (e) The phase transition QSH-CBI with the electric field U , where (C, C_s) labels with charge-Chern number C and spin-Chern number C_s . The variation of (f) the Hall conductivity σ_{xy} and (g) the ratio of spin polarization $|\delta S_{\uparrow}/\delta S_{\downarrow}|$ with U , respectively. We choose $u_F = 0.02$ eV.

To understand the physics of phase transition exhibited in Figs. 2(a)–(d), we make the Fourier transformation of Eq. (1) into the momentum space and expand it at the two Dirac valleys $\mathbf{K}_{\eta}$ ($\eta = \pm$) in the Brillouin zone (BZ). The resulting low-energy k -linear Hamiltonian near two Dirac valleys $\mathbf{K}_{\eta}$ reads

$$H_{\eta} = \hbar v_F (k_x \tau_x + \eta k_y \tau_y) + \eta t_{so} \sigma_z \tau_z - U \tau_z - \eta \hbar v_F^2 t_{\Omega} \tau_z, \quad (7)$$

where $\eta = \pm 1$ is the valley index, $v_F = \frac{\sqrt{3}}{2} a t$ (a is the lattice constant) is the Fermi velocity, and $\tau_{x,y,z}$ is the Pauli matrices in sublattice space. By diagonalizing the Hamiltonian H_{η} , an energy band gap with magnitude $2|\Delta E_{gap}|$ at $\mathbf{K}_{\eta}$ point reads

$$\Delta E_{gap} = \eta s_z t_{so} - U - \eta t_{\Omega}, \quad (8)$$

with $s_z = \pm 1$ for spin- $\uparrow$ and spin- $\downarrow$, respectively. The energy gap can be opened or closed by controlling both the light and electric fields. In contrast to the intrinsic spin-orbital coupling t_{so} which is spin- and valley-dependent, a circularly polarized light induces a new SOI term ηt_{Ω} , which is valley-dependent but not spin-dependent. The interplay between them causes different responses of spin-up and spin-down bands to the illumination parameter t_{Ω} , leading to the so-called type-II QAH, which has been proposed recently in an antiferromagnetic monolayer FeSe [44].

All phases can be characterized with topological invariant qualities (C, C_s) , where C and C_s are the well-known charge-Chern and spin-Chern number, respectively, calculated with the integral of a closed path $C_s = \frac{1}{2\pi} \sum_n \int_{BZ} d\mathbf{k} \Omega_{xy}^n(\mathbf{k})$ over the Berry curvature Ω_{xy}^n of the n -th band [37,45], where $\Omega_{xy}^n(\mathbf{k}) = -2\text{Im} \sum_{m(\neq n)} \frac{\langle n | \frac{\partial H}{\partial k_x} | m \rangle \langle m | \frac{\partial H}{\partial k_y} | n \rangle}{(E_m - E_n)^2}$. As a result, one can calculate the two invariants through $C = C_{\uparrow} + C_{\downarrow}$ and $C_s = (C_{\uparrow} - C_{\downarrow})/2$ with the spin-dependent Chern number C_{s_z} of each spin sector

$$C_{s_z} = \sum_{\eta=\pm} \frac{\eta}{2} \text{sgn}(\Delta E_{gap}). \quad (9)$$

In Fig. 2(i), we plot the phase diagram as a function of the illumination parameter t_{Ω} with $U = 0$, corresponding to the Figs. 2(a)–(d). Obviously, an off-resonant circularly polarized light can drive a QSH state with topological invariant qualities $(0, 1)$ into a QAH state with high Chen number $(2, 0)$. During this process, the light-induced effective spin-orbital coupling splits the spin-degenerate levels. With the enhancement of t_{Ω} , while the gap of spin-down band increases monotonously, the one of spin-up band reduces to zero first and then increases. As a consequence, the spin-dependent Chen number changes from $C_{\uparrow} = -1$ to $C_{\uparrow} = 1$ for the spin-up band whereas $C_{\downarrow} = -1$ remains unchanged for the spin-down band. The critical point $t_{\Omega} = \pm t_{so}$ is the point of exotic QSH-to-type-II QAH phase transition. We calculate the quantized Hall conductivity σ_{xy} and present it in Fig. 2(j). One can find that $\sigma_{xy} = 0$ for the helical edge of the QSH and $\sigma_{xy} = 2$ for the chiral edge of the QAH insulators, and between them a sharp jump at the critical point $t_{\Omega} = \pm t_{so}$ divides two different phases. Even so, the quantized conductivity σ_{xy} cannot characterize the anisotropy between different spin velocities owing to the constant conductance in one phase. Therefore, the analysis of other transport properties is needed. Since the edge state emerges in the gap, we pay our attention to the spin polarization in the low-energy regime. Thus, we further calculate the spin polarization $\delta S_z = \delta S_{\uparrow} - \delta S_{\downarrow}$ and plot ratio $|\delta S_{\uparrow}/\delta S_{\downarrow}|$ in Fig. 2(k). Interestingly, there appears a prominent signature of dip in $|\delta S_{\uparrow}/\delta S_{\downarrow}|$, which is exactly located at the QSH-QAH phase transition point. At the same time, the non-quantized δS_z exhibits the anisotropy between two spin velocities. The dip with $|\delta S_{\uparrow}/\delta S_{\downarrow}| = 0$ in Fig. 2(k) implies that only one spin species contributes to the transport. Deviating from the dip, both spin channels contribute to the transport but with different velocity, which is manifested by $0 < |\delta S_{\uparrow}/\delta S_{\downarrow}| < 1$. And, $|\delta S_{\uparrow}/\delta S_{\downarrow}| = 1$ indicates that the velocities of two spin species become equal in deep QSH or QAH states. At the same time, the signs of spin polarization in QSH and QAH phases are opposite.

From Eqs. (8) and (9), one can find the Chern number or topological phase also can be tuned with an external electric field U . Figs. 3(a)–(d) show the change of the band structure with the elec-

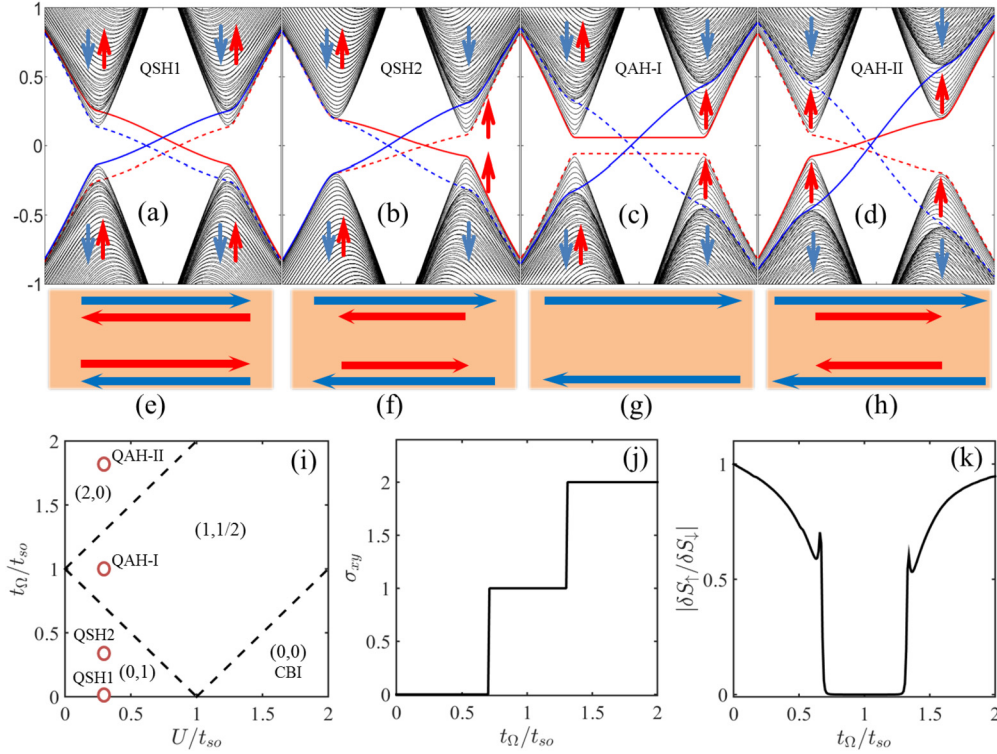


Fig. 4. (Color online.) (a)–(d) The evolution of energy band in a silicene nanoribbon with different light intensities ($t_\Omega = 0, 0.3, 1, 1.7$) and finite $U = 0.3$, in unit of t_{so} . The solid lines stand for up-edge state and the dashed lines for bottom-edge state. (e)–(h) Sketch map for edge states, corresponding to (a)–(d), respectively. The arrow direction denotes the transport direction and the arrow length stands for the velocity size. In (a)–(h), all red (blue) lines or arrows stand for spin-up (spin-down) electrons. (i) The phase diagram of dual-parameter (U, t_Ω), where (C, C_s) labels with charge-Chern number C and spin-Chern number C_s . The variation of (j) the Hall conductivity σ_{xy} and (k) the ratio of spin polarization $|\delta S_\uparrow/\delta S_\downarrow|$ with t_Ω , respectively. We choose $u_F = 0.02$ eV.

tric field strength U for the light field $t_\Omega = 0$. Different from the light field, the electric field breaks the inversion symmetry and U enters the energy gap with spin- and valley-independence, see Eq. (8). The interplay between the intrinsic SOI and U makes the spin-dependent gap move along opposite direction in each valley but remains the time reversal symmetry between valleys K and K' . Notice that the two spin-polarized edge channels connect the tips of the Dirac cones with the same spin between valleys K and K' . Therefore, the two edge states always keep the same amplitude of velocity with opposite sign under the protection of time-reversal symmetry. As a consequence, no anisotropy of transports is presented in Figs. 3(a)–(d), distinguishing from the anisotropic behaviors in Fig. 2. With the increase of U , the energy gap becomes narrower and narrower and the conducting and valence bands touch at a critical point $U = \pm t_{so}$. After that, the edge states vanish, which indicates the system entering the CBI states. The topological invariant qualities are given in Fig. 3(e), where the increase of U drives the QSH phase (0, 1) into the CBI phase (0, 0). Although there is no signature in the Hall conductivity due to $\sigma_{xy} = 0$ in Fig. 3(f), the finite spin polarization emerges, as shown in Fig. 3(g), which exhibits a jump of the spin polarization ratio $|\delta S_\uparrow/\delta S_\downarrow|$ in phase boundary between CBI and QSH phases. In the QSH phase, the ratio $|\delta S_\uparrow/\delta S_\downarrow| = 1$ is reached due to the isotropic velocity for different spins. Beyond the Hall conductivity, this jump behavior in the spin polarization ratio can be used to characterize the phase transition between CBI and QSH states.

When both light and electric fields are applied to the system, the energy gap is spin- and valley-dependent. The phase diagram in the dual-parameters (U, t_Ω) space is plotted in Fig. 4(i). The band structures at several typical states (denoted by circles in Fig. 4(i)) are presented in Figs. 4(a)–(d), where the asymmetry band structure between K and K' enhances the anisotropy of two edge-state velocities. Quantum phase transitions are realized by closing

and reopening the band gap with different parameters (U, t_Ω), and different phase sections are divided by the indicated straight lines. Compared with Fig. 2(i) or Fig. 3(e), between QSH and QAH phases or between QSH and CBI phases there appears a new topological phase, characterized with the topological invariants (1, 1/2) or $C_\uparrow = 0$ and $C_\downarrow = 1$. This is the typical type-I QAH state with only one edge state, which is spin polarized completely. Due to one channel, there emerges another quantized conductance e^2/h between 0 and $2e^2/h$, as shown in Figs. 4(j). The spin polarization from QSH to type-I QAH and type-II QAH is shown in Fig. 4(k). Obviously, the ratio $|\delta S_\uparrow/\delta S_\downarrow|$ exhibits different characters in different phases (notice that spin polarizations in QSH and type-II QAH phases have the same magnitude but with opposite sign). Similar to the spin polarization in Fig. 2(k), the slow change shows the anisotropic edge bands (different velocities for different spin species), and the ratio $|\delta S_\uparrow/\delta S_\downarrow|$ vanishes due to the annihilation of spin-up edge state in type-I QAH state.

4. Conclusion

We have in detail studied the anisotropic behavior of edge states in silicene nanoribbon due to topological phase transition tuned by both electric field and light irradiation. Due to the hexagonal lattice and buckled structure of silicene materials, the circularly polarized light causes different behaviors for spin-up and spin-down bands. As a consequence, a circularly polarized light induces a type-II QAH phase, manifested with the high Chern number and the strong anisotropic edge states. When a perpendicular electric field is exerted further, there exhibit intricate topological phases: QSH, type-I QAH, type-II QAH, and CBI. These topological phase transitions were also investigated in Ref. [21], where quantized topological invariants can characterize the different phases, but it fails for anisotropy of edge modes and

corresponding spin structure. In our study, we characterize the anisotropy of edge states beyond the quantized Hall conductivity and find that the spin polarization exhibits more signatures about the behavior of edge states, especially for the anisotropy transporting of spin-up and spin-down. Our results should be also suitable for other silicene-like materials such as stanene [46] or germanene [47,48]. The spin polarization can be detected with the spin- and angle-resolved photoemission spectroscopy (spin-resolved ARPES) [49,50], as well as the scanning Kerr microscopy measurement [51].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grants No. 11874016, the Key Program for Guangdong NSF of China under Grant No. 2017B030311003, and GDUPS (2017).

References

- [1] H. Li, X.H. Wang, A. Manchon, Valley-dependent spin-orbit torques in two-dimensional hexagonal crystals, *Phys. Rev. B* 93 (2016) 035417.
- [2] A. Dyrdak, J. Barnaś, Current-induced spin polarization and spin-orbit torque in graphene, *Phys. Rev. B* 92 (2015) 165404.
- [3] A.H. Castro Neto, F. Guinea, N.M.R. Peres, K.S. Novoselov, A.K. Geim, The electronic properties of graphene, *Rev. Mod. Phys.* 81 (2009) 109.
- [4] P. Vogt, P.D. Padova, C. Quaresima, J. Avila, et al., Silicene: compelling experimental evidence for graphenelike two-dimensional silicon, *Phys. Rev. Lett.* 108 (2012) 155501.
- [5] A. Fleurence, R. Friedlein, T. Ozaki, H. Kawai, Y. Wang, Y. Yamada-Takamura, Experimental evidence for epitaxial silicene on diboride thin films, *Phys. Rev. Lett.* 108 (2012) 245501.
- [6] L. Chen, Y.G. Yao, K.H. Wu, et al., Evidence for Dirac fermions in a honeycomb lattice based on silicon, *Phys. Rev. Lett.* 109 (2012) 056804.
- [7] N.D. Drummond, V. Zolyomi, V.I. Fal'ko, Electrically tunable band gap in silicene, *Phys. Rev. B* 85 (2012) 075423.
- [8] M. Ezawa, Quantized conductance and field-effect topological quantum transistor in silicene nanoribbons, *Appl. Phys. Lett.* 102 (2013) 172103.
- [9] X.-T. An, Y.-Y. Zhang, J.-J. Liu, S.-S. Li, Spin-polarized current induced by a local exchange field in a silicene nanoribbon, *New J. Phys.* 14 (2012) 083039.
- [10] K. Zborecki, R. Swirkowicz, J. Barnaś, Spin effects in thermoelectric properties of Al- and P-doped zigzag silicene nanoribbons, *Phys. Rev. B* 89 (2014) 165419.
- [11] C. Núñez, F. Domínguez-Adame, P.A. Orellana, L. Rosales, R.A. Römer, Silicene-based spin-filter device: impact of random vacancies, *2D Mater.* 3 (2016) 025006.
- [12] M. Ezawa, A topological insulator and helical zero mode in silicene under an inhomogeneous electric field, *New J. Phys.* 14 (2012) 033003.
- [13] K. Takeda, K. Shiraishi, Theoretical possibility of stage corrugation in Si and Ge analogs of graphite, *Phys. Rev. B* 50 (1994) 14916.
- [14] C.-C. Liu, W.X. Feng, Y.G. Yao, Quantum spin Hall effect in silicene and two-dimensional germanium, *Phys. Rev. Lett.* 107 (2011) 076802.
- [15] M. Ezawa, Valley-polarized metals and quantum anomalous Hall effect in silicene, *Phys. Rev. Lett.* 109 (2012) 055502.
- [16] G.G. Guzmán-Verri, L.C. Lew Yan Voon, Electronic structure of silicon-based nanostructures, *Phys. Rev. B* 76 (2007) 075131.
- [17] S. Cahangirov, M. Topsakal, E. Aktürk, H. Şahin, S. Ciraci, Two- and one-dimensional honeycomb structures of silicon and germanium, *Phys. Rev. Lett.* 102 (2009) 236804.
- [18] C. Kamal, A. Chakrabarti, M. Ezawa, Aluminene as highly hole-doped graphene, *New J. Phys.* 17 (2015) 083014.
- [19] A. Castellanos-Gomez, Why all the fuss about 2D semiconductors?, *Nat. Photonics* 10 (2016) 202.
- [20] M. Ezawa, Spin-valley optical selection rule and strong circular dichroism in silicene, *Phys. Rev. B* 86 (2012) 161407(R).
- [21] M. Ezawa, Photoinduced topological phase transition and a single Dirac-cone state in silicene, *Phys. Rev. Lett.* 110 (2013) 026603.
- [22] T. Oka, H. Aoki, Photovoltaic Hall effect in graphene, *Phys. Rev. B* 79 (2009) 081406(R).
- [23] G. Usaj, P.M. Perez-Piskunow, L.E.F. Foa Torres, C.A. Balseiro, Irradiated graphene as a tunable Floquet topological insulator, *Phys. Rev. B* 90 (2014) 115423.
- [24] Z.H. Gu, H.A. Fertig, D.P. Arovas, A. Auerbach, Floquet spectrum and transport through an irradiated graphene ribbon, *Phys. Rev. Lett.* 107 (2011) 216601.
- [25] N.H. Lindner, G. Refael, V. Galitski, Floquet topological insulator in semiconductor quantum wells, *Nat. Phys.* 7 (2011) 490.
- [26] J. Cayssol, B. Dóra, F. Simon, R. Moessner, Floquet topological insulators, *Phys. Status Solidi* 7 (2013) 101.
- [27] T. Kitagawa, T. Oka, A. Brataas, L. Fu, E. Demler, Transport properties of nonequilibrium systems under the application of light: photoinduced quantum Hall insulators without Landau levels, *Phys. Rev. B* 84 (2011) 235108.
- [28] T. Kitagawa, E. Berg, M. Rudner, E. Demler, Topological characterization of periodically driven quantum systems, *Phys. Rev. B* 82 (2010) 235114.
- [29] Y.F. Xu, X.F. Zhou, G.J. Jin, Detecting topological phases in silicene by anomalous Nernst effect, *Appl. Phys. Lett.* 108 (2016) 203104.
- [30] M. Ezawa, Topological phase transition and electrically tunable diamagnetism in silicene, *Eur. Phys. J. B* 85 (2012) 363.
- [31] H.-J. Duan, C. Wang, S.-H. Zheng, R.-Q. Wang, D.-R. Pan, M. Yang, Bulk RKKY signatures of topological phase transition in silicene, *Sci. Rep.* 8 (2018) 6185.
- [32] A. Dyrdak, J. Barnaś, V.K. Dugaev, Current-induced spin polarization in graphene due to Rashba spin-orbit interaction, *Phys. Rev. B* 89 (2014) 075422.
- [33] H. Li, A. Manchon, Tunable spin-charge conversion through topological phase transitions in zigzag nanoribbons, *Phys. Rev. B* 93 (2016) 235317.
- [34] X.F. Zhou, G.J. Jin, Silicene-based π and φ_0 Josephson junctions, *Phys. Rev. B* 95 (2017) 195419.
- [35] F.D.M. Haldane, Model for a quantum Hall effect without Landau levels: condensed-matter realization of the "parity anomaly", *Phys. Rev. Lett.* 61 (1988) 2015.
- [36] C.L. Kane, E.J. Mele, Quantum spin Hall effect in graphene, *Phys. Rev. Lett.* 95 (2005) 226801.
- [37] C.L. Kane, E.J. Mele, Z_2 topological order and the quantum spin Hall effect, *Phys. Rev. Lett.* 95 (2005) 146802.
- [38] X.F. Zhou, Y.F. Xu, G.J. Jin, Anomalous thermomagnetic effects in an epitaxial and irradiated graphene monolayer, *Phys. Rev. B* 92 (2015) 235436.
- [39] A. López, A. Scholz, B. Santos, J. Schliemann, Photoinduced pseudospin effects in silicene beyond the off-resonant condition, *Phys. Rev. B* 91 (2015) 125105.
- [40] J.-Y. Li, R.-Q. Wang, M.-X. Deng, M. Yang, In-plane magnetization effect on current-induced spin-orbit torque in a ferromagnet/topological insulator bilayer with hexagonal warping, *Phys. Rev. B* 99 (2019) 155139.
- [41] P.B. Ndiaye, C.A. Akosa, M.H. Fischer, A. Vaezi, E.-A. Kim, A. Manchon, Dirac spin-orbit torques and charge pumping at the surface of topological insulators, *Phys. Rev. B* 96 (2017) 014408.
- [42] A. Dyrdak, J. Barnaś, V.K. Dugaev, Current-induced spin polarization of a magnetized two-dimensional electron gas with Rashba spin-orbit interaction, *Phys. Rev. B* 95 (2017) 245302.
- [43] R. Yu, W. Zhang, H.-J. Zhang, S.-C. Zhang, X. Dai, Z. Fang, Quantized anomalous Hall effect in magnetic topological insulators, *Science* 329 (2010) 61; C.-Z. Chang, J. Zhang, X. Feng, J. Shen, et al., Experimental observation of the quantum anomalous Hall effect in a magnetic topological insulator, *Science* 340 (2013) 167.
- [44] Z.F. Wang, Z. Liu, J.L. Yang, F. Liu, Light-induced type-II band inversion and quantum anomalous Hall state in monolayer FeSe, *Phys. Rev. Lett.* 120 (2018) 156406.
- [45] K.S. Novoselov, E. McCann, S.V. Morozov, et al., Unconventional quantum Hall effect and Berry's phase of 2π in bilayer graphene, *Nat. Phys.* 2 (2006) 177.
- [46] F.-F. Zhu, W.-J. Chen, Y. Xu, C.-L. Gao, D.-D. Guan, C.-H. Liu, D. Qian, S.-C. Zhang, J.-F. Jia, Epitaxial growth of two-dimensional stanene, *Nat. Mater.* 14 (2015) 1020; A. Molle, J. Goldberger, M. Houssa, Y. Xu, S.-C. Zhang, D. Akinwande, Buckled two-dimensional Xene sheets, *Nat. Mater.* 16 (2017) 163.
- [47] M. Derivaz, D. Dentel, R. Stephan, M.-C. Hanf, A. Mehdaoui, P. Sonnet, C. Pirri, Continuous germanene layer on Al(111), *Nano Lett.* 15 (2015) 2510.
- [48] M.E. Dávila, L. Xian, S. Cahangirov, A. Rubio, G. Le Lay, Germanene: a novel two-dimensional germanium allotrope akin to graphene and silicene, *New J. Phys.* 16 (2014) 095002.
- [49] R. Raue, H. Hopster, R. Clauber, Observation of spin-split electronic states in solids by energy-, angle-, and spin-resolved photoemission, *Phys. Rev. Lett.* 50 (1983) 1623.
- [50] C.N. Veenstra, Z.-H. Zhu, M. Raichle, B.M. Ludbrook, A. Nicolaou, B. Slomski, G. Landolt, S. Kittaka, Y. Maeno, J.H. Dil, I.S. Elfimov, M.W. Haverkort, A. Damascelli, Spin-orbital entanglement and the breakdown of singlets and triplets in Sr_2RuO_4 revealed by spin- and angle-resolved photoemission spectroscopy, *Phys. Rev. Lett.* 112 (2014) 127002.
- [51] S. Matsuzaka, Y. Ohno, H. Ohno, Electron density dependence of the spin Hall effect in GaAs probed by scanning Kerr rotation microscopy, *Phys. Rev. B* 80 (2009) 241305(R).